

THIAGO SEIKI KATO

Full Stack Developer · AI Solutions Architect · Supply Chain & Data

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SUMMARY

Full stack developer and AI solutions architect, with a prior career in strategy consulting and operations management (EY, Webb, Rio 2016 Olympic Games). I work in the gap between business rule and implementation: I gather the requirement with the business area, model the data, and write the system — with no translation loss between the person who knows the rule and the person who writes the code.

Portfolio of 12 applications in production, pilot, or active development, built in Python/FastAPI and TypeScript/React on PostgreSQL, Supabase, and Google Cloud. Domains span catalog standardization with embeddings and vector search, discrete-event simulation for industrial operations, a tax calculation engine with decimal precision, and retrieval-augmented generation with citations traceable back to source.

Hands-on multi-LLM orchestration (Gemini, Claude, OpenAI, Ollama), with close attention to per-call cost, multi-tenant isolation enforced through Row Level Security, and workflow design that keeps the final decision with a human wherever an error is expensive.

TECHNICAL SKILLS

- **Languages:** Python 3.12, TypeScript, SQL, JavaScript
- **Backend:** FastAPI, Pydantic, SQLAlchemy, Alembic, Celery, Redis, pytest, WebSockets, REST APIs
- **Frontend:** React 18/19, Next.js 14/16, Vite, Tailwind CSS, React Flow, TanStack Query, Flutter
- **Data:** PostgreSQL, Supabase, Firestore, Row Level Security, pgvector, pg_cron, Pandas, Polars, NumPy, BigQuery
- **AI & ML:** Gemini, Claude, OpenAI, Ollama, LangChain, RAG, embeddings, Vertex AI Vector Search, hybrid search (RRF), A2A protocol, XGBoost, scikit-learn, TensorFlow
- **Cloud & Infrastructure:** Google Cloud Run, Cloud Tasks, Cloud Functions, Cloud Scheduler, Azure, Docker, Vercel, Git
- **Simulation & Statistics:** SimPy, NetworkX, Monte Carlo, ARIMA/SARIMA, statsmodels, SciPy, stress testing
- **Business:** Supply chain, S&OP, demand planning, strategic sourcing, Brazilian tax regime, Power BI, Tableau

SELECTED PROJECTS

Supply Chain, Simulation and Decision Support

Supply Chain Pipeline Builder | *Personal project · Actively evolving*

Python · FastAPI · React · Pandas · statsmodels · SciPy

Visual builder for supply chain analysis and demand forecasting pipelines: each card is an operation, available columns are detected from the preceding card, and every step previews its own output.

- 15 ETL operations, inventory analysis (ABC/XYZ classification, safety stock, reorder point, economic order quantity), and forecasting via moving averages and ARIMA/SARIMA, with MAE, RMSE, and MAPE accuracy metrics.
- Monte Carlo simulation across 10,000+ iterations for cost-optimized safety stock sizing.
- Any pipeline exports as a runnable Python script or Jupyter notebook, generated entirely in the browser — a deliberate product decision so the analyst's work is never locked inside the tool.
- Sole ownership declared in the repository LICENSE. Currently being integrated with a corporate demand planning system.

OmniTwin | *Personal project · In development*

Python 3.11 · SimPy · NetworkX · FastAPI · Pydantic · React Flow · MQTT · Celery · Redis

Digital twin for industrial operations resilience, built on the premise that operations do not fail at the average — they fail at the combination of events.

- Discrete-event simulation engine in SimPy with stochastic failure via MTBF/MTTR, changeover setup matrix by SKU, and bill-of-materials validation at every processing step.
- Calculation engine fully decoupled from the modeling interface through strict JSON schemas validated with Pydantic: replacing the frontend never touches the engine, and changing the engine never invalidates a saved model.
- Stress testing across a swept parameter (e.g. supplier delay), returning inventory level series, failure-and-repair Gantt per asset, the node that most constrained flow, and the financial impact of unmet demand.
- Live telemetry over MQTT with an explicit fallback rule: any parameter out of range or unavailable reverts to the model's stochastic value.

Enterprise Systems and Business Verticals

Match Hub | *Personal project, subsequently adopted into operations · In production*

Python 3.12 · FastAPI · Cloud Run · Firestore · Vertex AI Vector Search · Gemini Flash · React 19 · Cloud Tasks

AI-driven standardization and matching of medical and hospital supply items, solving the problem of the same product being described dozens of different ways across hospitals, distributors, and manufacturers — which makes reliable price comparison and demand consolidation impossible.

- LLM normalization, 768-dimension embeddings, and vector search, followed by a second LLM re-ranking pass; ambiguous cases route to a human curation queue rather than becoming a silent bad match.
- Multi-tenant with per-API-key isolation and a central canonical catalog with a propose-and-approve workflow; asynchronous imports via Cloud Tasks in batches of 5,000 items without blocking the API.
- Promoting items between catalogs reuses already-processed descriptions, embeddings, and categories — a caching architecture designed around LLM cost, so the same item is never paid for twice.
- Two Cloud Run services deployed from a single image, separated by environment variable: one serves requests, the other consumes the queue.

Tax-Hub | *Enterprise system · Complete MVP*

TypeScript · React 19 · Vite · Tailwind CSS 4 · Supabase · PostgreSQL · decimal.js · Row Level Security

Tax simulation for interstate pricing in Brazil, where a single sale triggers several interacting taxes (ICMS-ST, DIFAL, FCP, IPI, PIS/COFINS) and the order in which they enter the calculation changes the result. The harder problem is not complexity but rate of change: new state legislation means a new formula.

- Formulas live outside the codebase: the tax analyst registers a rule, validates it against known golden cases, and publishes — legislative change stopped being an engineering ticket waiting in a deploy queue.
- Custom expression language with hand-written tokenizer, parser, and evaluator, deliberately avoiding eval, so a user-authored formula is never executed blindly.
- Pure calculation engine with no I/O and no framework dependency, using decimal precision arithmetic; step-by-step calculation trace exportable to Excel with live formulas or printed as an audit document.
- Reverse solver by bisection for target-price and target-margin modes; tax rule resolver by scope, specificity, and effective-date window.

LexMind | *Client engagement · First vertical in development*

Next.js 16 · Supabase · PostgreSQL 17 · pgvector · Claude Sonnet · Gemini 2.5 Pro · Vitest · Playwright

Legal copilot for public tenders and government contracting under Brazil's procurement law (Law 14.133/2021), designed around the premise that an assistant which fabricates case law is worse than no assistant at all.

- Every statement the copilot makes carries a source identifier resolved in real time against a verified document: no source, no claim.
- Hybrid retrieval combining vector and full-text search through Reciprocal Rank Fusion, because exact legal terminology and semantic similarity fail in opposite situations.
- An analyst agent reads the full tender notice and returns a structured diagnosis — qualification requirements, deadlines, restrictive clauses, internal contradictions — before the lawyer opens the file.
- Tenant isolation between law firms tested as code, with a dedicated test suite covering Row Level Security policies.

Additional systems in production or under contract

- **Multi-Agent Assistant** — *Supply4Med · in operation*. Natural-language operations copilot inside Microsoft Teams, reading the Odoo ERP directly through its API, with domain-specialist agents (CRM, projects, support) and a plan-then-act cycle for multi-step questions. Python, LLM, Azure, RAG.
- **Condominium Procurement** — *client engagement · in use*. End-to-end purchasing workflow for a property management company, from quotation request to payment confirmation, with a Kanban board by stage, a supplier comparison matrix exportable to Excel and PDF, line-item award, and SLA alerts via pg_cron with daily deduplication. Python 3.12, FastAPI, Supabase, React 18, TypeScript.
- **Lótus Scout Group Management System** — *personal project · pro bono · in production*. Financial and educational management for a 460-member volunteer organization, with compound-rule billing, expense reimbursement workflow with value-based approval tiers, and monthly closing locked after board approval. Runs at zero operating cost on free tiers, with a keep-alive mechanism preventing database auto-pause. Next.js 14, Supabase, Cloud Run.

R&D in Multi-Agent Architecture

A personal lab for validating AI architectures under real conditions before proposing them in a corporate setting — focused on integration cost, autonomy between components, and the practical limits of each approach.

JARVIS Ecosystem | *Personal project · Full loop running in production*

Python 3.12 · FastAPI · Gemini · Pydantic · SQLAlchemy · Alembic · PostgreSQL · argon2 · pytest

- Four agents across four repositories and four processes, communicating over the A2A protocol on HTTP, with no agent able to reach another's database. Orchestrator (JARVIS), asset memory (Alfred), finance with its own web dashboard (Lucius), and identity/session extracted into an installable package shared by both frontends (Portaria).
- Automated tests replace all three LLM endpoints with deterministic doubles by explicit decision — a test cannot depend on an API key or on a response that varies; live validation against the real model runs separately, with agents online simultaneously.

- Every query exposes drill-down to the rows behind each figure and explicitly declares what was excluded from the total and how much it represents; actuals and forecasts are held in separate collections so no consumer can sum them by accident.
- Transferable finding: agents separated by protocol cost more to build and return genuine autonomy — including the ability for one to fail without taking down the others.

Additional lab experiments

- **CyberSec Cockpit** — *personal project · in development*. Security testing orchestration under an authorization-first rule: the authorization document is audited by AI and the scope extracted from it governs what may be executed. Chains static, dependency, dynamic, and network analysis into a single pipeline, with human review before report generation. Celery, Redis, WebSockets, LLM provider routing by cost and task fit.
- **Detective AI** — *personal project · working prototype*. Assessment of prompt engineering and critical thinking through an investigative simulation, with LLM-driven suspects that remember, lie, and contradict themselves. Bring-your-own-key architecture with the option of fully local execution via Ollama, so no data leaves the infrastructure. Flutter, FastAPI, Docker.
- **Dedé** — *personal project · v1*. Adaptive study platform for children, with three role-specific agents and adaptation logic deliberately kept outside the LLM — an exponential moving average weighted by difficulty, in plain testable code — for behavior that is predictable, auditable, and cheap. AI provider switchable through an admin panel with no redeploy. React, Firebase, Cloud Functions v2.

PROFESSIONAL EXPERIENCE

Supply4Med | Head of Intelligence · Jul 2024 – Present

- Founded and led the Intelligence function, delivering the AI agent ecosystem integrating Google Gemini, Microsoft Teams, Outlook, Odoo ERP, and scheduled WebJobs, enabling business users to query project, CRM, and task data in natural language.
- Built an AI solution for item description standardization and purchase demand consolidation, raising catalog data quality to the point where reliable price comparison and consumption analysis became possible.
- Implemented a retrieval-augmented (RAG) document intelligence platform for centralized internal document search, plus AI agents for automated market intelligence gathering supporting commercial prospecting.
- Designed a data governance model based on per-profile tool sets and an access hierarchy, alongside autonomous agents generating and delivering proactive operational reports.
- Led the definition and instrumentation of the company's strategic KPIs, aligning metrics to business objectives through to development readiness.
- **Previously Head of Operations and Control:** achieved a 66% reduction in process execution time through automation, with Power BI order management dashboards.

Advison Consulting | Specialist Advisor · Jun 2023 – Mar 2024

- **LeadTime project (Grupo Boticário, major Brazilian cosmetics group):** analysis of operational data extracted from SAP and delivery of reports and dashboards driving end-to-end lead time reduction, using GCP/BigQuery, SQL Server, Dataprep, Power BI, and Tableau.
- **ScoreCredit project:** predictive credit risk model for a pharmaceutical distributor, applying statistical and machine learning techniques (RFV) to optimize credit limits and expand sales under controlled risk.

EzTools | Project Manager · May 2022 – Jun 2023

- Solutions architect and project manager for multiple RPA and BPMS implementations, focused on designing and optimizing complex business processes for automation. Clients: Eurofarma, Atento, Sicoob.

What's on Tap? | Founder · Oct 2016 – May 2022

- Founded and ran a food industry startup, with full ownership of the business plan, cash flow, budgeting, and supplier relationships.

Rio 2016 — Olympic Games Organizing Committee | Procurement Manager · Aug 2015 – Sep 2016

- Strategic sourcing and acquisition of high-volume categories — utilities, transportation for staff, athletes and spectators, and Olympic Village equipment — serving on the purchase approval committee.

EY | Project Manager · Oct 2012 – May 2015

- Restructured the procurement function of a Brazilian company operating in Venezuela through category management and strategic planning; delivered strategic sourcing projects for consumer goods, chemicals, and agribusiness clients.
- Established a PMO governing demand generation, procurement, and logistics pre-projects for a large wholesaler, and formulated go-to-market strategy for an instant meals product line.

Webb / BrainNet | Project Manager · Sep 2007 – Oct 2012

- Delivered logistics solutions for e-commerce and pharmaceutical clients, including a performance-based compensation model for logistics providers; strategic sourcing across shopping mall services, heavy agricultural equipment, education, and retail.
- Built the market entry business plan for a German heavy equipment manufacturer entering Brazil, and led a pioneering study on biomass as an alternative energy source for the beverage industry.

EDUCATION AND LANGUAGES

- **MBA, Strategic and Economic Project Management** — Fundação Getulio Vargas (FGV), São Paulo, 2013
- **BSc, Business Administration** — FGV-EAESP, São Paulo, 2003
- **Languages:** Portuguese (native) · English (fluent, TOEIC 940) · Spanish (advanced)

CONTINUING EDUCATION

- Data Science & Supply Chain Analytics — expected completion Jan 2026
- LLM Applications using LangChain — Udemy, 2024 · [certificate](#)
- Python — Hashtag Treinamentos, 2024 · [certificate](#)
- Data Science (statistics, NumPy, scikit-learn, TensorFlow 2.0) — 365 Careers, 2024 · [certificate](#)
- R Programming for Statistics and Data Science — 365 Careers, 2024 · [certificate](#)
- SQL — Hashtag Treinamentos, 2022 · [certificate](#)
- Agile Project Management with Scrum — Udemy, 2020 · [certificate](#)
- Partner at Impact Hub São Paulo and Impact Hub Florianópolis; investor in Impact Hub Belo Horizonte